30561 THUREL Jules

PSI

Feedback control of a Buck converter to stabilise the output voltage from a variable input source.

The World Bank's Lighting Global programme promotes access to solar energy for off-grid populations through the use of solar kits. For the 733 million people without electricity, these stand-alone photovoltaic systems are an excellent alternative.

My project deals with the feedback control of a Buck converter, a key component of these systems. The challenge is to design a regulation loop capable of stabilising the output voltage despite variations in sunlight and of maintaining high efficiency.

The candidate certifies having worked alone.

Thematic positioning (STEP 1):

- *INDUSTRIAL SCIENCES (Control engineering)*
- *INDUSTRIAL SCIENCES (Electrical engineering)*
- *INDUSTRIAL SCIENCES (Energy engineering)*

Keywords (STEP 1):

Keywords (in French)	**Keywords (in English)**
Convertisseur Buck	*Buck converter*
Asservissement	*Feedback control*
Rapport cyclique	*Duty cycle*
Correcteur PI	*PI controller*
Panneaux solaires	*Solar panels*

Annotated bibliography

According to data from the International Energy Agency (IEA), about 730 million people still have no access to a reliable source of electricity [3]. These figures underline the importance of a decentralised energy source such as a stand-alone solar installation. To increase the overall efficiency of the photovoltaic system, an efficient means of converting energy is necessary. Moreover, the voltage supplied by solar panels is too high (12-24 V) for direct use by

electronic devices. To meet these requirements, many switching converter topologies exist. Among them, the DC-DC Buck converter is a suitable solution [1]. It both converts energy with an efficiency above 90 % and lowers the output voltage, which then becomes usable by other devices.

The operation of the Buck converter, documented in the Unitrode design seminars [5], relies on the fast switching of an electronic switch (MOSFET) at a high frequency (fs = 200 kHz), making it possible to regulate the average output voltage through the duty cycle D according to the relation: V_out = D × V_in. The coefficient D corresponds to the duty cycle of the PWM (Pulse Width Modulation) component, an element that generates a square-wave signal with variable duty cycle in order to drive the switch [2]. Concretely, when the switch is closed, the source delivers energy to the load, while also storing energy in the Buck's internal storage elements (capacitor and inductor). These storage elements also supply energy to the load when the switch is open. Thus, energy transfer is maintained between the source and the customer, while making it possible to choose the level of the delivered voltage. Furthermore, these elements filter the voltage of the switching cell, keeping a smoothed average voltage for the user [1].

To implement this system, several elements require particular attention. Firstly, the use of an inductive component imposes current continuity in the loop even if the switch is open. Here, a freewheeling diode performs this function. Secondly, certain criteria must be met for the converter to work properly and to be approximated by a linear system. These criteria are: the continuous conduction mode (CCM), the small-signal assumption and the cut-off frequency of the filter. The continuous conduction mode corresponds to the case where the current in the inductor never drops to zero [2][5]. Sizing the components, for a given switching frequency, is essential to satisfy this criterion. The second point is based on the work of Middlebrook [6], which makes it possible to linearise the Buck equations with a Taylor expansion limited to first order. The last criterion is to have a cut-off frequency of the LC filter that is small compared with the switching frequency: fc << fs [1].

Understanding how the Buck converter works was necessary to tackle its feedback control. At the heart of my project, this voltage regulation loop makes it possible to maintain a stable output despite fluctuations in sunlight. To design this feedback control, the converter is modelled in the frequency domain by its small-signal transfer function. This function is obtained by linearising its equations around a steady-state operating point. This approach is valid for disturbances of small amplitude and of frequency well below the switching frequency [6]. Consequently, classical control engineering tools such as Bode diagrams and the Nyquist criterion can be used to ensure a stable, fast and accurate regulation loop. These various requirements can be validated by choosing a suitable compensator such as a PI or a PID [4]. The parameters of these compensators can be chosen with the Ziegler and Nichols method as a first approach [7], then refined using simulations and a study of the open loop of the system [1].

Research question retained

Faced with variations in the voltage delivered by a photovoltaic panel, to what extent does the feedback control of a Buck converter constitute a relevant solution for guaranteeing a stable and regulated output voltage?

Objectives of the candidate's TIPE

- Model the Buck converter in LTspice with a view to sizing the components.
- Study and model the feedback loop (choice of compensator).
- Build the converter experimentally on a prototyping board (breadboard).
- Control the converter and study its step response.
- Assess the robustness of the system and conclude on its suitability for the needs of stand-alone photovoltaic installations.

Bibliographic references (STEP 1)

[**1**] RIDLEY, RAYMOND B. : Power Supply Design. Volume 1 : Control. *: Ridley Engineering, Inc., 2017*

[**2**] DURAND, CLÉMENT. : Conception et intégration d'un convertisseur buck en technologie 28 nm CMOS orientée plateformes mobiles *: Université de Lyon, École Doctorale Électronique, Électrotechnique, Automatique, 2018 : https://theses.hal.science/tel-02918212v1/file/these.pdf*

[**3**] AGENCE INTERNATIONALE DE L'ÉNERGIE (AIE) : World Energy Outlook 2025. *: AIE Publications, 2025 : https://www.iea.org/reports/world-energy-outlook-2025*

[**4**] RAHIMI, AMIR M., PARTO, PARVIZ, ASADI, PEYMAN : Compensator Design Procedure for Buck Converter with Voltage-Mode Error-Amplifier (Application Note AN-1162) *: International Rectifier (Infineon Technologies), 2013.*

[**5**] UNITRODE CORPORATION. : Power Supply Design Seminar Manual – SEM-1000. *: Texas Instruments, 1996.*

[**6**] MIDDLEBROOK, R. D., & ČUK, S. : A General Unified Approach to Modelling Switching Converter Power Stages. *: Proc. IEEE PESC, 1976, p. 73-76 : https://www.mikrocontroller.net/attachment/124600/SSA_Cuk_Middlebrook.pdf*

[**7**] ZIEGLER, JOHN G., ET NICHOLS, NATHANIEL B. : Optimum Settings for Automatic Controllers. *: Trans. ASME, vol. 64, 1942, p. 759-768.*

DOT

[1] *: June 2025 – Research on the Buck converter and theoretical study of its operation.*

[2] *: Summer 2025 – Getting to grips with the LTspice simulation software and further literature research on the Buck and its voltage feedback control.*

[3] *: September 2025 – First conclusive simulations allowing correct sizing of the components, in order then to order the first essential components: MOSFET, resistors and capacitors*

[4] *: November 2025 – Finalisation of the two LTspice models, and contact with a technician in a laboratory for the design of the inductor and advice (high-side driver) for the breadboard implementation*

[5] *: December 2025 / February 2026 – Construction of several sub-circuits on breadboard: bootstrap circuit of the driver, modification and adaptation of the PWM for voltage-mode operation, and assembly of the Buck's power stage.*

[6] *: March 2026 – First test of the feedback control, showing a lot of noise because of the power stage (high-frequency switching, impact of the switching on the parasitic elements of the breadboard circuit). "Saturation" phenomena (loss of controllability between the main regulation op-amp and the PWM) were observed and resolved in order to stay within the operating margins.*

[7] *: May 2026 - Complete and satisfactory assembly of the feedback control, allowing measurements of the system in the time domain (step response) and in the frequency domain using a Transfer Function Analyzer.*

[8] *: End of May 2026 – Test in real conditions with a photovoltaic panel and a DC motor at the output.*